Molecular Neuroscience and Synaptic Transmission in Traumatic Stress

Chapter 5

In this chapter

Learning objectives 2
Introduction 2
5.1 Molecular organisation of synapses 3
5.2 Electrical and chemical signalling 3
5.3 Vesicle release 4
5.4 Receptor families 4
5.5 Second-messenger systems 5
5.6 Gene transcription 5
5.7 Protein synthesis 6
5.8 Oxidative stress 6
5.9 Mitochondrial regulation 7
5.10 Molecular adaptation after stress 7
Synthesis 7
Composite case study 8
Review and discussion questions 8
Research assignments 9
Chapter summary 9
References 9

Central question. How do molecules, receptors and gene expression produce the synaptic changes associated with threat learning, and what can human studies infer about them?

Learning objectives

- describe the molecular architecture of the synapse and the cycle of neurotransmitter release;
- distinguish ionotropic from metabotropic receptors and explain second-messenger signalling;
- explain how experience alters gene transcription and protein synthesis;
- evaluate the roles of oxidative stress and mitochondrial function;
- distinguish molecular mechanism from molecular speculation in trauma research.

Introduction

Synapses are where computation becomes chemistry. A presynaptic terminal converts an electrical signal into vesicle fusion and neurotransmitter release; the postsynaptic membrane converts that chemical signal back into electrical or biochemical change. Around this core, an elaborate machinery of scaffolding proteins, receptors, kinases and transcription factors determines whether a synapse strengthens, weakens or persists.

Trauma research has borrowed heavily from this machinery. Long-term potentiation and long-term depression provide models of learning; reconsolidation research suggests memories can become labile; stress hormones alter receptor trafficking and gene expression. These are genuine mechanisms, studied in detail. The interpretive challenge is that most of the detailed work is in reduced preparations and animals, and the gap to a human symptom is large.

This chapter describes the molecular events most relevant to traumatic stress, with attention to what each finding does and does not establish.

> **A note on terminology.** *"Molecular" findings often refer to mRNA or protein levels in tissue, not to activity in a living human brain. A change in receptor expression after stress is a real biological observation; it is not a measurement of a thought, a feeling or a diagnosis.*

5.1 Molecular organisation of synapses

A chemical synapse has three components: the presynaptic terminal, the synaptic cleft (about 20 nanometres wide), and the postsynaptic density. The terminal contains synaptic vesicles filled with neurotransmitter, anchored to the cytoskeleton and poised for release. The active zone is the site of fusion, organised by proteins including RIM, Munc13 and the SNARE complex.

The postsynaptic density is a dense protein network containing receptors, scaffolding molecules such as PSD-95, signalling enzymes and cytoskeletal linkers. Its composition determines which receptors are present, how they cluster, and how they respond to activity. Changes in the postsynaptic density are one mechanism by which synaptic strength is regulated.

Astrocytic processes often surround the synapse, forming the tripartite arrangement described in Chapter 4. They remove neurotransmitter, buffer ions and release modulators, and thus shape the signal that the postsynaptic cell receives.

5.2 Electrical and chemical signalling

Signals travel within neurons electrically and between neurons chemically. An action potential arriving at the terminal depolarises it, opens voltage-gated calcium channels, and triggers vesicle fusion. Neurotransmitter diffuses across the cleft and binds to receptors on the postsynaptic membrane.

Ionotropic receptors are ligand-gated ion channels: binding opens a pore and produces a fast electrical response lasting milliseconds. Metabotropic receptors are G-protein-coupled and produce slower, longer-lasting effects through second messengers. The same neurotransmitter can act through both types; glutamate, for example, acts at ionotropic AMPA and NMDA receptors and at metabotropic glutamate receptors.

This dual organisation allows a single transmitter to serve fast signalling and slow modulation. It also explains why drugs targeting a transmitter system can have diverse effects: blocking a receptor may alter rapid transmission, sustained excitability, or both.

5.3 Vesicle release

Synaptic vesicles are filled with neurotransmitter by specific transporters. They dock at the active zone, become primed, and fuse with the membrane in response to calcium entry. Fusion is mediated by SNARE proteins—syntaxin, SNAP-25 and synaptobrevin—which form a complex that pulls the vesicle and plasma membranes together.

Release has multiple modes. Synchronous release produces fast transmission. Asynchronous release, which occurs after the initial calcium transient, can prolong signalling. Spontaneous release occurs without action potentials and contributes to the maintenance of synaptic connections.

After fusion, vesicle membrane is retrieved by endocytosis and recycled. This cycle must be tightly regulated, because sustained high-frequency firing would otherwise deplete the terminal. Toxins that cleave SNARE proteins—botulinum and tetanus toxins—abolish release, which is why they are both dangerous and clinically useful.

5.4 Receptor families

Glutamate receptors mediate most excitatory transmission. AMPA receptors produce fast depolarisation and are the workhorses of synaptic transmission. NMDA receptors are permeable to calcium and are blocked by magnesium at resting potential; they open when the membrane is depolarised and glutamate is bound, making them coincidence detectors. This property underlies their role in long-term potentiation. Kainate receptors modulate transmission and excitability.

GABA receptors mediate most fast inhibition. GABA-A receptors are chloride channels modulated by benzodiazepines, barbiturates and alcohol. GABA-B receptors are metabotropic and produce slower inhibition through potassium channels.

Other families matter for trauma. Noradrenergic alpha-1 and beta receptors mediate arousal and are implicated in stress effects on prefrontal cortex. Dopamine D1 receptors modulate working memory in an inverted-U fashion, so that both too little and too much signalling impair performance (Arnsten, 2009). Serotonin receptors modulate mood, sleep and appetite, though the relationship between serotonin and depression is more complex than early theories suggested. Cannabinoid CB1 receptors regulate neurotransmitter release and are involved in fear extinction and stress coping.

5.5 Second-messenger systems

Metabotropic receptors activate intracellular cascades. A common pathway involves G proteins that stimulate adenylate cyclase, producing cyclic AMP, which activates protein kinase A. Another involves phospholipase C, producing inositol trisphosphate and diacylglycerol, which mobilise calcium and activate protein kinase C.

These cascades amplify signals and can produce lasting change by phosphorylating receptors, ion channels and transcription factors. The cAMP response element-binding protein (CREB) is a key node: phosphorylation of CREB recruits co-activators and drives transcription of genes involved in synaptic plasticity, including BDNF.

Stress engages these pathways directly. High catecholamine levels during stress activate cAMP and protein kinase C signalling in prefrontal cortex, weakening network connectivity and impairing working memory (Arnsten, 2009). The same signalling can strengthen amygdala function. This is a molecular account of the shift from reflective to reflexive control under threat.

5.6 Gene transcription

Experience changes gene expression. Transcription factors such as CREB, NF-κB and the glucocorticoid receptor bind to regulatory regions and alter the rate at which genes are transcribed into mRNA. This is how a transient signal can produce a durable change.

Epigenetic mechanisms regulate access to DNA. DNA methylation typically represses transcription when it occurs in gene promoters; histone acetylation generally increases it. These marks can be altered by experience, including stress, and can persist for some time. The most discussed example in trauma research is FKBP5, a co-chaperone of the glucocorticoid receptor. Early-life stress has been associated with altered FKBP5 methylation and with altered stress reactivity in adulthood.

The evidence is genuine but limited. Human epigenetic studies often use peripheral tissue—blood or saliva—because brain tissue is rarely available. Whether peripheral marks reflect brain marks is uncertain. Effects are small, samples are modest, and replication is incomplete. Claims that trauma is "written into the genome" overstate what has been shown.

5.7 Protein synthesis

Long-lasting synaptic change requires new protein synthesis. Blocking protein synthesis shortly after learning prevents the consolidation of long-term memory in many paradigms. This is the basis of the reconsolidation literature: after retrieval, a memory may become temporarily labile and require new protein synthesis to be restabilised (Nader et al., 2000).

Reconsolidation has attracted clinical interest because it suggests a window in which a traumatic memory could be modified. The evidence in humans is mixed. Some studies report that administering a beta-blocker or performing extinction training during the reconsolidation window reduces fear responses; others fail to replicate, and the boundary conditions—timing, strength of reactivation, age of memory—are not settled (Vervliet et al., 2013).

Protein synthesis is also required for the structural changes that accompany learning: new dendritic spines, altered receptor composition, and changes in presynaptic release probability. These changes are not permanent; they turn over, which is why maintenance matters.

5.8 Oxidative stress

Reactive oxygen species are normal products of metabolism and participate in signalling. When production exceeds antioxidant capacity, oxidative stress damages lipids, proteins and DNA, and can trigger cell death. The brain is vulnerable because of its high metabolic rate, high lipid content and relatively modest antioxidant defences.

Stress and inflammation can increase oxidative stress. In animal models, chronic stress is associated with oxidative damage in the hippocampus and prefrontal cortex, and antioxidant treatment can attenuate some behavioural effects. In humans, markers of oxidative stress are elevated in some psychiatric conditions, but the findings are non-specific and affected by diet, smoking, exercise and medication.

Claims that antioxidant supplements treat trauma-related conditions are not supported by adequate trials. The biological rationale is plausible; the clinical evidence is not there.

5.9 Mitochondrial regulation

Mitochondria generate ATP, buffer calcium, regulate apoptosis and participate in signalling. Neurons have high energy demands, particularly at synapses, and mitochondria are transported to sites of high activity. Mitochondrial dysfunction impairs synaptic function and increases vulnerability to excitotoxicity.

Chronic stress affects mitochondrial function. Glucocorticoids alter mitochondrial metabolism, and inflammatory mediators impair oxidative phosphorylation. In some studies, mitochondrial DNA copy number and respiratory capacity differ in people with stress-related conditions, but these findings are preliminary and methodologically demanding.

Mitochondria also participate in the response to psychological stress through their role in cortisol synthesis in the adrenal cortex, which is a reminder that the relevant biology is distributed across organs, not confined to the brain.

5.10 Molecular adaptation after stress

The molecular response to stress is adaptive in the short term and potentially costly when sustained. Acute catecholamine release sharpens attention and mobilises energy. Acute glucocorticoid release supports glucose availability and consolidates emotionally significant memory. Inflammatory signalling recruits immune defences.

When these systems remain activated, the consequences include impaired prefrontal function, altered hippocampal structure, increased amygdala reactivity, sleep disruption, insulin resistance and impaired immune function. This is the molecular version of allostatic load (McEwen, 1998).

The clinical implication is that interventions which restore the conditions for regulation—sleep, safety, exercise, social connection—act on the same systems that stress has perturbed. That does not make them "molecular treatments"; it means that molecular biology and everyday life are connected.

Synthesis

Table 5.1 summarises molecular processes and their relevance to traumatic stress.

Table 5.1. Molecular processes and trauma-relevant significance

Process	Function	Trauma relevance	Interpretive caution
Receptor trafficking	Changes synaptic strength	Fear learning, extinction	Measured mostly in animals
Second messengers	Amplify and integrate signals	Stress effects on prefrontal cortex	Inverted-U dose effects
Transcription	Converts transient signals to lasting change	Consolidation, adaptation	Peripheral tissue limits in humans
Protein synthesis	Stabilises long-term change	Reconsolidation window	Boundary conditions unresolved
Oxidative stress	Damage when antioxidants overwhelmed	Hippocampal vulnerability	Non-specific markers
Mitochondrial function	Energy supply, calcium buffering	Synaptic resilience	Difficult to measure in vivo

Composite case study

A patient asks whether a supplement will "repair" the molecular damage caused by childhood abuse. A careful answer distinguishes what is known—chronic stress affects inflammatory, endocrine and synaptic systems—from what is not: no supplement has been shown to reverse trauma-related changes or to treat PTSD. The clinician offers evidence-based treatment and explains that sleep, exercise and social support have measurable biological effects without being "cures".

Review and discussion questions

Why is the NMDA receptor described as a coincidence detector, and why does this matter for learning?

How does the inverted-U relationship between catecholamines and prefrontal function change the way we think about "stress hormones"?

What are the limits of inferring brain epigenetics from blood samples?

Why has reconsolidation been difficult to translate into reliable clinical treatment?

Distinguish a marker of oxidative stress from a mechanism of harm.

Why should molecular findings not be described as measuring thoughts or feelings?

Research assignments

- Design a study that could test whether a psychotherapy changes a molecular marker, specifying the tissue, timing and confounding variables.
- Critically appraise one human epigenetic study of childhood adversity, noting sample size, tissue, replication and effect size.
- Write a one-page briefing for clinicians on what "molecular" evidence does and does not mean for treatment decisions.

Chapter summary

Synaptic transmission depends on vesicle release, receptor activation and intracellular signalling. Metabotropic pathways and transcription factors convert brief signals into lasting change, and protein synthesis stabilises long-term memory. Stress engages these systems through catecholamines, glucocorticoids and inflammatory mediators, producing adaptive changes that can become costly when sustained. The evidence is strongest in animal and cellular preparations; human studies rely on indirect markers. Molecular knowledge should sharpen both mechanism and humility.

References

Arnsten, A. F. T. (2009). Stress signalling pathways that impair prefrontal cortex structure and function. *Nature Reviews Neuroscience, 10*(6), 410–422. https://doi.org/10.1038/nrn2648

McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *New England Journal of Medicine, 338*(3), 171–179. https://doi.org/10.1056/NEJM199801153380307

Nader, K., Schafe, G. E., & LeDoux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. *Nature, 406*(6797), 722–726. https://doi.org/10.1038/35021052

Vervliet, B., Craske, M. G., & Hermans, D. (2013). Fear extinction and relapse: State of the art. *Annual Review of Clinical Psychology, 9*, 215–248. https://doi.org/10.1146/annurev-clinpsy-050212-185542